

DEPARTMENT OF COMPUTER SCIENCE

GUJARAT UNIVERSITY

MCA-2

SUBJECT : Operating Systems – Practical

Assignment 2

1. Write a script that takes file name from user and display all line starting with a or b or c. (Use grep/sed)
2. Write a script that takes file name from user and display all line starting not with a or b or c. (Use grep/sed)
3. Write a script that takes file name from user and display all line starting not with a or b or c with line numbers. (Use grep/sed)
4. Write a script that takes file name from user and substitute all spaces “ “ with # value. (Use grep/sed)
5. Write a script that takes file name from user and display all line start with t or T and second character must be either ‘h’ or ‘s’. (Use grep/sed)
6. Write a script that takes file name from user and display all line start with space(‘ ’) . (Use grep/sed)
7. Write a script which takes input from a file, with multiple records, as:
Firstname:middlename:lastname:address:city:pin:phone
and displays output as:
Record 1
Lastname middlename firstname
Address

City – pin
Phone

Record 2
Lastname middlename firstname
Address
City – pin
Phone
and so on, for all records.

8. Write a script that shows usernames and no. of processes running for them.
9. The book master file contains the fields book_no, book_name, author, dateofpurchase ,each field is separated by hypen. write a script for
 - (A) Add
 - (b) Modify
 - (C) Deletefrom above file.
10. Write a shell script to display list of files which can be either regular or directory along with number of links in ascending order of links.
11. write a script to count number of vowels in file irrespective of case.
12. write a script that accepts a string followed by one or more file names from command line and display no of lines that consists of given string each file.
13. Create a text file with the headings and data as bill_no, cust_no, cust_name, address, city, pin, current_meter_reading, previous_meter_reading, month. Write a script that takes the input from this file and displays the bill for the query against cust_no/bill_no/cust_name. (input at least 15/20 records in this text file); otherwise message that no record exists.
14. Write a shell script to generate summary from the sales.dat file.Sales made by 3 salesman by selling 3 products are

entered in a file. Add atleast 10 records. The format is as shown below:

Salesman:Product1:Product2:Product3

Sample data:

Mr. Abhishek Sharma:21:29:12

Mr. Akash Srivastava:11:15:28

Mr. Abhilash Dwivedi:31:04:13

Calculate the followings :

1. Total sales of the company
2. Highest sold product
3. Best salesman (who sold the most)
4. Worst salesman (who sold the least)

15. Write a shell script to generate summary from a file :

“student.dat” with following format :

Student_no : student_name : gender : marks1 : marks2 :
marks3

Each field must be separated by a delimiter ‘-‘

Process the following queries:

1. Calculate the total marks of each student
2. Calculate the percentage of marks for each student
3. Count the total number of male and female students
4. Count the total number of students who pass and those who fail.

16. Create a file employee.dat with fields:

emp_id:emp_name:department:basic_salary:hra:da:pf

(Insert at least 15 records)

Write a shell script to:

Calculate gross salary (basic + hra + da)

Calculate net salary (gross - pf)

Display details of employee based on emp_id / emp_name

Display highest paid employee

17. Electricity Bill Summary

Create a file electricity.dat with fields:

consumer_no:name:units_consumed:month

Write a shell script to:

Calculate bill using slab:

0-100 → ₹2/unit

101-200 → ₹3/unit

Above 200 → ₹5/unit

Display total revenue collected

Find consumer with highest bill

Search record by consumer_no/name

18. Library Management System

Create a file library.dat with fields:

book_id:title:author:student_name:issue_date:return_date

Write a shell script to:

Display all issued books

Find books issued to a particular student_name

Count total issued books

Display overdue books (assume current date input)

19. Product Inventory Analysis

Create a file inventory.dat with fields:

product_id:product_name:quantity:price

Write a shell script to:

Calculate total inventory value

Find product with highest quantity

Find product with lowest price

Search product by product_id/name

20. Cricket Score Analysis

Create a file cricket.dat with fields:

player_name:runs:balls:fours:sixes

Write a shell script to:

Calculate strike rate of each player

Find highest scorer

Count players scoring above 50

Display summary of all players